

Project Title: Beyond Binary Detection: Explainable AI (XAI) Deepfake Localization via Frequency-Aware Segmentation

1. Introduction & Problem Definition

The Concept "Beyond Binary" is an advanced deepfake detection framework that moves past simple "Real vs. Fake" labels to provide **interpretable forensic evidence**. Instead of just looking at pixels, this project interrogates the hidden mathematical structure of an image.

The Strategy We use a **Dual-Stream approach**:

- 1. **Spatial Stream:** Analyzes visible textures and shapes (what the eye sees).
- 2. **Frequency Stream:** Uses **Discrete Cosine Transform (DCT)** to detect invisible "checkerboard artifacts" and spectral noise, the digital fingerprints left behind by AI generation math.

The Goal: By fusing these two streams, the system identifies forgeries that are visually perfect but mathematically inconsistent. Using **Explainable AI (XAI)**, we generate heatmaps that pinpoint the manipulated areas, turning a "Black Box" prediction into a transparent forensic tool.

2. The Problem & Our Proposed Solution

The Current Problem (The "Gap")	Our Solution (The "Beyond Binary" Approach)
Spectral Blindness: Standard AI looks at pixels. Modern fakes are pixel-perfect, making visual detection unreliable.	Frequency-Aware Analysis: We use DCT (Discrete Cosine Transform) to look at the "mathematical noise" that humans and standard AI miss.

The "Black Box" Issue: Most detectors just give a "Real" or "Fake" label with no explanation, which isn't useful for forensics.	Explainable AI (XAI): We integrate Grad-CAM to produce heatmaps that highlight the exact area of the forgery.
Storage Bottlenecks: High-end datasets are too large (100GB+) for solo researchers with limited cloud storage (15GB).	Decoupled Data Pipeline: We engineered a "Stream-and-Clean" script that processes data locally without clogging permanent storage.
Overfitting: Models often only catch the specific type of fake they were trained on (e.g., only faces).	Cross-Domain Benchmarking: We test against multiple generators (Diffusion + GANs) to ensure the model catches "AI logic," not just images.

3. Primary Research Question

"Can combining image math (Frequency) with visual pixels (Spatial) create a more explainable and accurate deepfake detector?"

Sub-Questions (The 3 Goals)

1. **The Accuracy Question:** Can the model find "hidden" AI artifacts that the human eye cannot see?
2. **The Transparency Question:** Can we use heatmaps to show *where* a photo was manipulated instead of just saying "Fake"?
3. **The Efficiency Question:** Can high-end AI research be done effectively using limited (15GB) cloud storage?

4. Dataset Information:

1. Overview of Data Sources

To achieve a robust, "Beyond Binary" detection system, this research utilizes a **Multi-Phase Benchmark Strategy**. We leverage three distinct datasets that represent different eras and methods of AI generation:

- **CIFAKE (Primary Baseline):** A dataset containing 120,000 images (60,000 Real / 60,000 Synthetic). The synthetic class is generated via **Stable Diffusion v1.4**. This serves as the foundation for identifying general diffusion-based artifacts.

Link: [CIFAKE: Real and AI-Generated Synthetic Images | Kaggle](#)

- **FaceForensics++ (Forensic Benchmark):** A standard in the facial manipulation field. We utilize the **C32 (Compressed)** version to test the model's ability to detect fakes under real-world social media compression scenarios.

Link: [FaceForensics++ Dataset \(C23\)](#)

- **MS COCOAI (SOTA Stress Test):** A dataset featuring high-fidelity images of complex scenes and objects. This is used to validate the model's ability to **generalize** beyond simple objects to complex, state-of-the-art AI synthesis.

Link: [COCO 2017 Dataset | Kaggle](#)

2. Dataset Composition

Each image in the pipeline is processed through two distinct lenses:

1. **Spatial Domain:** Standard RGB pixel data used to identify visual inconsistencies (e.g., texture mismatches, lighting errors).
2. **Frequency Domain:** Extracted via **Discrete Cosine Transform (DCT)** to reveal high-frequency "Spectral Signatures" mathematical grid patterns left by AI upsampling that are invisible to the human eye.

3. Data Engineering & Pipeline Management

Given the large-scale nature of these datasets (exceeding 100GB in total) and the **15GB storage limitation** of standard cloud environments, we implemented a **Decoupled Ingestion Architecture**:

- **Kaggle API Integration:** Data is fetched dynamically from the Kaggle repository.
- **Ephemeral Local Workbench:** Large datasets are unzipped into the local Colab runtime disk (/content/datasets/). This provides high-speed read/write access and prevents storage overflow on Google Drive.
- **Permanent Artifact Syncing:** Only trained model weights, performance logs, and **XAI Heatmaps** are synced to permanent storage, ensuring research continuity without storage bottlenecks.

Dataset	Images	Target Manipulation	Purpose in Project
CIFAKE	120,000	Latent Diffusion	Baseline Training
FF++	5,000+ Frames	GAN / Face-Swap	Forensic Robustness
MS COCOAI	10,000+	SOTA Synthesis	Generalization Testing

5. Literature Review

GSheet Link:  DIP = DM

Current State of the Art (What Exists)

- **Single-Domain Analysis:** Most research focuses solely on the "Spatial Domain," training AI to look for visual flaws like blurring or irregular skin textures.

- **Binary Labeling:** Standard industry tools usually output a simple "Real" or "Fake" percentage, which provides no context for forensic investigations.
- **Dataset Overfitting:** Many existing models are trained on specific datasets (like faces only) and fail when shown general objects or landscapes created by new AI.
- **Infrastructure Dependency:** Academic deepfake research typically assumes the user has access to enterprise-grade servers with hundreds of gigabytes of storage.
- **Black-Box Logic:** High-accuracy models often lack transparency; they provide a result but cannot explain which specific part of the image triggered the "Fake" alert.

Innovation in "Beyond Binary" (What is New)

- **Dual-Stream Mathematical Fusion:** My work integrates a "Frequency Stream" that uses the Discrete Cosine Transform (DCT) to catch "Spectral Signatures" (math errors) that are invisible in standard pixel views.
- **Forensic Localization (Heatmapping):** Unlike binary systems, this project implements **Grad-CAM** to highlight the exact area of manipulation, turning a simple "Yes/No" into a visual piece of evidence.
- **Decoupled Engineering Strategy:** I have designed a "Stream-and-Clean" data pipeline that allows high-end research on 100GB+ datasets using a standard 15GB Google Drive.
- **Cross-Generator Validation:** This project specifically tests the model against multiple AI architectures (GANs vs. Diffusion) to prove it detects "AI logic" rather than just memorizing one type of image.
- **Human-in-the-Loop Interpretability:** By moving "Beyond Binary," the system is designed as a tool for human experts to verify AI decisions rather than just blindly trusting a computer's score.
- **Resource-Aware Optimization:** I have engineered the code to use **Transfer Learning**, which adapts powerful pre-trained "brains" to this forensic task in hours rather than weeks of training.

6. Expected Contributions:

The primary contribution of this work is the development of an efficient, interpretable, and frequency-aware framework for deepfake detection. By moving

"Beyond Binary" labels, this research provides a transparent forensic tool capable of operating under significant resource constraints.

The specific contributions are outlined below:

1. Dual-Stream Feature Fusion (Bridging the Spectral Gap)

We introduce a unified architecture that bridges the gap between spatial semantics and spectral anomalies. By synchronizing a **Frequency-aware Stream** with a traditional **Spatial Stream**, the framework captures **high-frequency periodic artifacts** (upsampling noise) that remain undetected by standard pixel-based models. This allows the system to identify forgeries that are visually perfect but mathematically inconsistent.

2. Resource-Efficient Data Engineering (Democratizing Research)

Recognizing the hardware and storage constraints of cloud-based research (15GB limit), I have engineered a **Decoupled Ingestion Pipeline**. By implementing a **"Stream-and-Clean"** local workbench strategy, this work demonstrates that forensic research on 100GB+ datasets is feasible within standard cloud tiers. This provides a blueprint for conducting high-fidelity AI research without requiring expensive, enterprise-grade server infrastructure.

3. Forensic Interpretability & Spatial Localization

This project facilitates a paradigm shift from simple binary classification to **Spatial Forensic Localization**. Through the integration of **Grad-CAM** (Gradient-weighted Class Activation Mapping), the system converts abstract AI probability scores into **verifiable visual evidence**. This turns the "Black Box" of AI into a transparent tool, allowing human investigators to pinpoint exactly *where* an image has been manipulated.

4. Cross-Domain Generalization (Detecting the "Math of the Machine")

Unlike models that overfit to specific datasets, this pipeline is designed for **Cross-Generator Validation**. By testing the system against divergent AI architectures (e.g., Latent Diffusion vs. GANs), the research validates the model's ability to identify the **fundamental mathematical fingerprints** of AI synthesis. This ensures the detector remains robust against new, unseen AI generators.

5. Functional Model Deployment & User Interface

Beyond the theoretical model, this project contributes a **functional forensic application**. By deploying the "Beyond Binary" engine via a web-based interface (e.g.,

Streamlit or Gradio), we provide an end-to-end tool where users can upload images and receive real-time results. This includes the final classification score and the localized Grad-CAM heatmap, bridging the gap between a "code-based model" and a **ready-to-use forensic utility**.

7. Support for State-of-the-Art (SOTA) AI Synthesis

As generative AI evolves, visual flaws (like skin textures or lighting) are being eliminated, leading to **Spectral Blindness** in traditional detectors. The "Beyond Binary" framework is specifically engineered to remain effective against SOTA synthesis (e.g., Stable Diffusion XL, Midjourney v6) through the following mechanisms:

1. Mathematical Invariance

While SOTA models produce visually perfect pixels, the underlying **upsampling math** remains consistent. By using the **Discrete Cosine Transform (DCT)**, our model ignores the "perfect" visual surface and targets the mathematical grid-patterns that SOTA models *must* create to generate high-resolution images.

2. Diverse Generator Benchmarking

We validate our framework against the **MS COCOAI** dataset. Unlike older benchmarks that focus on simple faces, MS COCOAI contains complex, SOTA-generated scenes and objects. This ensures our model identifies "AI Logic," the fundamental way machines calculate images rather than just memorizing old-fashioned "fake looks."

3. Resilience to High-Fidelity Synthesis

SOTA synthesis often hides its fingerprints in high-frequency noise. Our **Frequency Stream** is specifically tuned to analyze these high-frequency coefficients where modern "Denoising Diffusion" models leave their unique spectral signatures.

8. Proposed Methodology: The Beyond Binary Framework

The architecture is designed as a **Dual-Stream Fusion Network** that treats deepfake detection as a multi-modal signal processing task. Instead of relying on raw pixels alone, the system interrogates the image in two distinct domains simultaneously.

Phase 1: Pipeline Engineering & Decoupled Data Ingestion

- **Objective:** To bypass physical hardware constraints (15GB cloud limit) while processing high-fidelity datasets (100GB+).
- **Methodology:** Implementation of a **Decoupled Storage Architecture**.
 - **Dynamic Streaming:** Utilizing the Kaggle-python API to stream data directly into an **Ephemeral Local Workbench** (Colab T4 runtime).
 - **Non-Persistent Processing:** A "Stream-and-Clean" script unzips and processes batches in RAM/Local Disk, syncing only the **Model Weights (.h5/.keras)** and **Performance Logs** to Google Drive. This ensures zero storage-related downtime.

Phase 2: Dual-Stream Model Architecture

- **Objective:** To eliminate "Spectral Blindness" by fusing visual and mathematical features.
- **The Spatial Stream:** Uses a pre-trained **Xception backbone** utilizing **Depthwise Separable Convolutions**. This branch focuses on **Micro-Texture Analysis**, identifying high-frequency inconsistencies in lighting, shadows, and semantic boundaries. Unlike standard models, Xception's architecture is specifically tuned to catch the "point-like" artifacts (irregular eye reflections or mismatched skin pores) often left by GANs.
- **The Frequency Stream:** Implements a **Discrete Cosine Transform (DCT)** preprocessing layer. This branch converts the image into its spectral components to detect **Periodic Correlation Artifacts**. These "checkerboard" patterns are mathematical anomalies created by AI upsampling layers (Transposed Convolutions) that are invisible in the RGB domain.
- **Late Fusion Module:** Features from both streams are merged using **Vector Concatenation** following a **Global Average Pooling (GAP)** layer. By using a "Late Fusion" strategy, we allow each branch to reach high-level abstraction before the final **Dense classification head** makes the "Beyond Binary" decision.

Phase 3: Multi-Phase Training & Stress Testing

- **Objective:** To achieve cross-generator generalization (Diffusion vs. GAN).
- **Baseline Training:** Initial optimization using the **CIFAKE** dataset to learn the spectral signatures of Latent Diffusion Models (Stable Diffusion).
- **Domain Adaptation:** Fine-tuning on **FaceForensics++ (C32)**. The use of compressed (C32) data is critical to ensure the model remains robust against real-world social media compression algorithms.
- **Hyperparameter Optimization:** Implementing a **Learning Rate Scheduler** and **Dropout Layers (0.5)** to prevent the model from memorizing specific dataset backgrounds.

Phase 4: XAI Integration & Forensic Localization

- **Objective:** To transform the "Black Box" into an interpretable forensic tool.
- **Methodology:** Integration of **Grad-CAM (Gradient-weighted Class Activation Mapping)**.
- **Visual Evidence:** Instead of a binary score, the system generates **Spatial Heatmaps**. These heatmaps highlight the specific pixels that contributed most to the "Fake" classification. If an AI manipulated only the mouth or eyes, the Grad-CAM output will localize that specific region, providing verifiable evidence for forensic investigators.

Phase 5: Model Deployment & Prototype Testing

- **Objective:** To transition the trained model into a user-facing application.
- **Methodology:**
 - **Backend Integration:** Exporting the trained Dual-Stream weights into a lightweight inference engine.
 - **Interface Design:** Building a web dashboard using **Streamlit** to allow for easy image uploads and visual output display.
 - **Localization Overlay:** Engineering a real-time visualization layer that overlays the Grad-CAM heatmap directly onto the user's uploaded image.
- **Milestone:** A live, interactive prototype where a user can upload a file and see the "Forensic Proof" (Heatmap + Score) instantly.

9. Expected Technical Workflow

The "Beyond Binary" system operates through a structured 5-step data flow, moving from raw input to forensic visualization.

Step 1: Adaptive Data Ingestion

- **Action:** The system connects to the Kaggle API to stream the selected benchmark (CIFAKE/FF++).
- **Process:** Images are unzipped into the **Ephemeral Local Workbench** (RAM/Local Disk).
- **Efficiency:** The "Stream-and-Clean" logic ensures that only active training batches occupy space, preventing storage overflow.

Step 2: Dual-Stream Feature Extraction

- **Spatial Path:** The image is passed through the **Xception** backbone to identify RGB-level inconsistencies (textures/edges).
- **Frequency Path:** Simultaneously, a **DCT Transformation** is applied to the image to convert it into the spectral domain, exposing hidden mathematical grid artifacts.

Step 3: Feature Fusion & Classification

- **Action:** The high-level features from both streams are merged via **Late Fusion (Concatenation)**.
- **Decision:** The Dense layers of the network process the combined vector to produce a final "Probability of Synthesis" score.

Step 4: Forensic Localization (The XAI Layer)

- **Action:** If an image is flagged as "Fake," the **Grad-CAM engine** is triggered.
- **Process:** The system traces the "winning" neurons back to the spatial coordinates of the image to generate a **Localization Heatmap**.

Step 5: Interactive Deployment Output

- **Action:** The final result is pushed to the **Streamlit Web Interface**.
- **User View:** The investigator sees the original image, the mathematical confidence score, and the heatmap overlay identifying the specific region of manipulation.